CSCE 4813 – Computer Graphics
Midterm Exam – Spring 2018

Student Name: Solutions

Student UAID: _____

I pledge that I have neither given nor received unauthorized help on this exam

Instructions: This is closed book exam. Students are allowed one 8.5 x 11 page of notes, but no calculators or other electronic devices. The exam worth a total of 60 points. Please read all questions before starting the test and schedule your time accordingly.

1) Geometric Models and Transformations (20 points)

We have discussed rotation, translation and scaling transformations in class. For this question, we will be using the following notation:

- R(theta, ax, ay, az) denotes a rotation by theta degrees around the vector (ax, ay, az),
- T(dx, dy, dz) denotes a translation by (dx, dy, dz) units, and
- S(sx, sy, sz) denotes a scaling by (sx, sy, sz) units.

a) Assume that you have read the coordinates for your letter outline into three float arrays "X[10], Y[10], Z[10]" where the initial Z values are zero Describe the <u>sequence of step</u> you would use to transform this geometric model without changing the shape so it is centered at (0,0,0) and the height of the letter (size in Y direction) is 100. Be specific about which of the R, T, S operations you would perform, and how you would calculate the parameters to each of these functions.

- loop over 3 arrays to find Xmin, Xmax, Ymin, Ymax, Zmin, Zmax.
- translate by $\left(-\left(\frac{X_{max}+X_{min}}{2}\right), -\left(\frac{Y_{max}+Y_{min}}{2}\right), -\left(\frac{Z_{max}+Z_{min}}{2}\right)\right)$
- scale by $\left(\frac{100}{Y_{max}-Y_{min}}\right)$ in all 3 dimensions.
- no rotation is needed.

b) What is the formula for <u>translating</u> a point by (dx, dy, dz) units. Show how this can be implemented using a 4x4 matrix multiplication.

$$x' = x + T_x$$
$$y' = y + T_y$$
$$z' = z + T_z$$

$$(x'\ y'\ z'\ 1) = \begin{pmatrix} 1 & 0 & 0 & T_x \\ 0 & 1 & 0 & T_y \\ 0 & 0 & 1 & T_z \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \\ 1 \end{pmatrix}$$

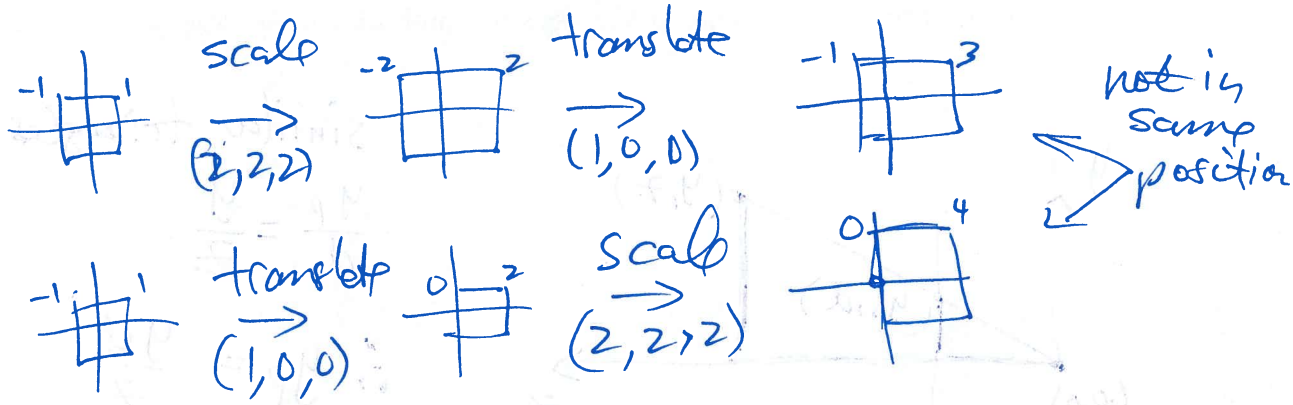
c) What is the formula for <u>scaling</u> a point by (sx, sy, sz) units. Show how this can be implemented using a 4x4 matrix multiplication.

$$x' = x \cdot S_x \qquad y' = y \cdot S_y \qquad z' = z \cdot S_z$$

$$(x'\ y'\ z'\ 1) = \begin{pmatrix} S_x & 0 & 0 & 0 \\ 0 & S_y & 0 & 0 \\ 0 & 0 & S_z & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \\ 1 \end{pmatrix}$$

d) Explain with formulas and/or diagrams why you think that <u>order normally matters</u> when you perform translation and scale transformations. Can you think of a special case where order does not matter and ST = TS?

- consider box centered at origin



- special case when scale = 1, translate = 0.

2) Computer Viewing (20 points)

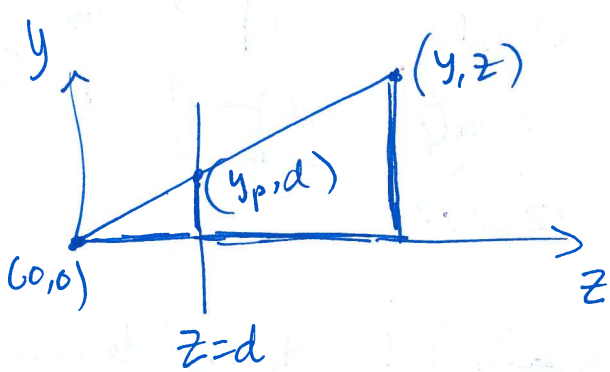
The goal of perspective projection is to transform three-dimensional object coordinates (X,Y,Z) into two-dimensional screen coordinates (xp, yp). To do this we typically put the center of projection is at (0,0,0) and the projection plane at Z=d.

a) What are the <u>formulas</u> for calculating (xp, yp) in terms of (X,Y,Z) and d?

$$x_p = \frac{X}{Z/d} = \frac{Xd}{Z}$$

$$y_p = \frac{y}{Z/d} = \frac{yd}{Z}$$

b) Draw a "side view" diagram of perspective projection to illustrate <u>one</u> of these formulas.

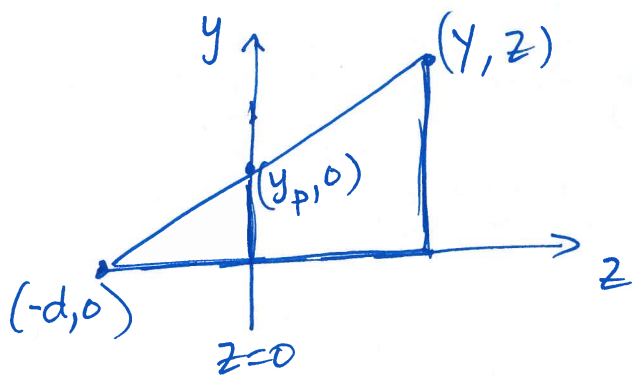


similar triangles.

$$\frac{y_p}{d} = \frac{y}{z}$$

$$\therefore y_p = \frac{yd}{z}$$

c) Some graphics systems move the center of projection to (0,0,-d) and project the (X,Y,Z) coordinates of all points in the scene onto the plane Z=0. Will this <u>increase or decrease</u> the projected size of objects in the scene? Explain with diagrams or equations.



similar triangles

$$\frac{y_p}{d} = \frac{y}{z+d}$$

$$\therefore y_p = \frac{yd}{z+d}$$

- since $\frac{yd}{z+d}$ is smaller than $\frac{yd}{z}$ the projected size will decrease.

d) The default parallel projection for most graphics systems assumes that the projection plane is Z=d. What would the formulas be if we performed a <u>top view</u> projection onto the Y=d plane instead?

- default parallel projection

$$x_p = x, \quad y_p = y, \quad z_p = d$$

- top view projection

$$x_p = x, \quad y_p = d, \quad z_p = z$$

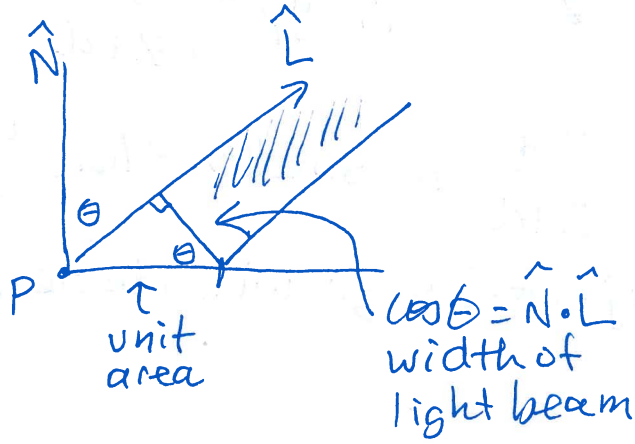
- to display object on screen z_p values would have to be shown on y-axis

3) Computer Shading (20 points)

In the Phong shading model, the amount of light reflected from a surface has three components: the diffuse term D, the specular term S, and the ambient term A.

Assume that the camera is located at location (0,0,0) and we are calculating the shading model of a point on a surface at (Px,Py,Pz). The surface normal is given by (Nx,Ny,Nz) and the direction from the point to the light source is given by (Lx,Ly,Lz).

a) Draw a diagram to illustrate how Phong calculates how much <u>light per unit area</u> reaches the point (Px,Py,Pz) on the surface? What is the formula used by Phong?



$$I = \underbrace{k_d \cdot I_d \cdot \hat{L}\cdot\hat{N}} + k_s \cdot I_s (\hat{V}\cdot\hat{R})^{\alpha} + k_a \cdot I_a$$

b) How is this light per unit area value used in the Phong shading model? Is this part of the diffuse term D, the specular term S, or the ambient term A?

- light per unit area is used to estimate how bright an object will appear if all that light scatters in all directions
- this is part of the diffuse term D in the phong model.

c) What pieces of information are needed to calculate the viewpoint vector (Vx,Vy,Vz) and the ideal reflection direction (Rx,Ry,Rz)? Describe in words how these two vectors are calculated.

- the viewpoint vector points from point P on object towards camera at (0,0,0)

$\vec{V} = -(P_x, P_y, P_z) / \|\vec{P}\|$

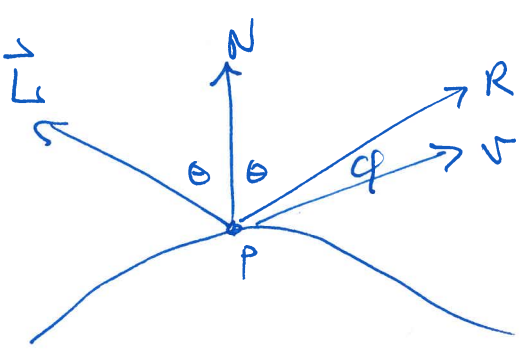
- the reflection direction is calculated using vector towards light $\vec{L}$ and surface normal $\vec{N}$

- we choose $\vec{R}$ so angle of incidence equals angle of reflection φ

$\vec{R} = 2 \cdot (\vec{L} \cdot \vec{N})\vec{N} - \vec{L}$ (then normalize to unit length)

d) How are the V and R vectors above used in the Phong shading model? Is this part of the diffuse term D, the specular term S, or the ambient term A?

- the $\vec{R}$ and $\vec{V}$ vectors are used to estimate how much light reflects directly off the surface and hits the camera



- this is part of the specular term S

- $k_s I_s \cdot (\vec{V} \cdot \vec{R})^{\alpha}$

where α controls how mirror like the surface is